Poroelastic rupture and cycles: implementation and verification plan

Biot poroelasticity in the dynamic, quasi-dynamic and fully dynamic solvers, behind an option that is off by default · revision 3, 20 Oct 2026

Revision 3 (Chunhui Zhao, 20 Oct 2026: "prepare for the next revision, and layout implementation plan carefully with sufficient unit tests to guard the accuracy, prepare the verification tests with SeisSol and own tests"). (1) The equations now live in one place, the derivation document revision 1.2; the derivations of revision 2.2 (its §8) are replaced by references to it. This plan uses F1 as decided and F2, F3 as recommended; both still need a decision (§1.2). (2) New §5 breaks the implementation into 28 reviewable steps, each with its files, functions and tests. (3) New §6 specifies 56 unit tests with expected values, tolerances and a matrix of which test guards which equation. (4) New §7 specifies the verification: 18 tests of our own with known answers, and five SeisSol comparisons, S0–S4, each with its pass test. Four new decisions, D9–D12, are in §1.2. (5) A new script, `poro_test_refs.py`, computes every reference value. It corrects three numbers of the revision 2.2 TPV205 table (fast P 6029, not 6051 m/s; slow P 979, not 1046 m/s; diffusivity 0.019, not 0.022 m²/s), which came from a stale $K_s = 60$ GPa in `biot.py`. (6) §9 lists every build and run with its estimated cost and the words that would approve it. Revision 2.2 is kept in `plan-2026-10-19/`. Branch `nitsche-dynamic-rupture-poroelastic` is at `5b2c77d`; no code has been written.

1. What this plan answers

We add Biot poroelasticity to the Nitsche dynamic-rupture solver without changing its character. The bulk stays a symmetric interior penalty (SIPG) DG method in second-order displacement form, and the fault stays the Nitsche split-node interface with slip as a prescribed field. The new unknown is the relative fluid displacement w, in the same DG space as the solid displacement u. With the total-stress split (F2), the solid block is exactly today's elastic operator built with the undrained modulus λ_u, and everything new is one divergence form used three times, an element-local mass and drag, a fitted Newmark coefficient for the fluid, and a fault pressure taken from the fluid block's own flux. Friction sees Terzaghi's effective normal stress. All new code goes in `miniapps/seas/multiphysics/poroelastic/`; a deck option, **off by default**, switches it on, and with the option off every existing run stays bit-identical.

The same bulk then carries over to the quasi-dynamic solver (inertia dropped, pressure diffusing implicitly in a split step around RK45) and to the fully dynamic cycle (the hand-off grows from (u, d, ψ) to (u, w, d, ψ)). The equations, their derivations and the code each one maps to are in the derivation document revision 1.2 (`poro-derivation-rev1.2.pdf`), cited below as "derivation (eq.)".

How accuracy is guarded. Four layers, each needed before the next is trusted:

1. **Nothing old changes.** With the option off, every existing Nitsche test passes and a TPV205 run is byte-identical to the trunk build (U0-1, U0-2). Where the poroelastic code reuses an elastic path, the reuse is checked bit for bit (U2-4, U2-15, U4-4).
2. **Every new function is tested at round-off** against an independent computation or an exact identity, with the expected values frozen in a committed reference file (§6).

3. **The discretization converges** at its design order against exact solutions: plane waves, patch tests, manufactured solutions, Terzaghi, Mandel and Cryer (§6, §7.1).
4. **The physics agrees** with independent results: our elastic TPV205 golds through the undrained-equivalent medium, and SeisSol run by us on the same meshes at the same orders (§7.2).

A tolerance is never loosened after a result is seen without Chunhui Zhao's approval, and a phase ends only when its own tests and every earlier test pass.

1.1 The plan in one table

Phase	What it adds	Steps (§5)	Tests that end it
P0	Trunk merge (D12), folders, the option (off), hooks, static background pore pressure in σ_n'	P0.0–P0.3	U0-1…U0-6
P1	Biot material sample, derived constants, dispersion, time-step bound	P1.1–P1.2	U1-1…U1-6
P2	Bulk (u, w) operator, sources, local mass and drag, stepper	P2.1–P2.6	U2-1…U2-19, V-B1–V-B3, V-B5
P3	Analytic 3D point sources	P3.1	V-B4
P4	Fault pressure, coupling traction, σ_n', outputs	P4.1–P4.3	U4-1…U4-10, V-F1–V-F4
P5	SeisSol builds and comparisons, `tpv5-poro`	P5.1–P5.3	U5-1…U5-3, S0–S2 (S3, S4 optional)
P6	Speed, sealed fault, user document	P6.1–P6.3	U6-1…U6-3
P7	Quasi-dynamic: split step, implicit diffusion	P7.1–P7.4	U7-1…U7-6, V-Q1–V-Q7
P8	Fully dynamic cycle: the hand-off	P8.1–P8.2	U8-1…U8-3, V-C1–V-C2

Each step ends with a review page and Chunhui Zhao's approval before its commit. Work stays in this branch and its one local repository (`/Users/chunhuizhao/projects/seas-mfem-poroelastic`); any new clone, checkout or worktree is asked for first. The Mac runs the unit tests (one run at a time, under the run lock, `OMP_NUM_THREADS=1`); the cloud thread does git; Vista builds and runs only from a pushed commit with a clean tree, under the CLAUDE.md guard, and every job is submitted only on Chunhui Zhao's words for that job (§9).

1.2 Decisions

D1–D8 were answered on 20 Oct 2026, all as recommended: full Biot (u, w) (D1); the fitted Newmark stepper (D2); a permeable fault first (D3); both rupture benchmarks (D4); the deck option as stated (D5); Vista for SeisSol and our code, on EAR26024 (D6); the quasi-dynamic split step (D7); MUMPS refactorization with PCG in between (D8). F1 (the injection rate Q as the general source, the `tpv5-poro` start as a special case) was decided the same day.

Six questions remain open. The recommendation in each row is what the rest of this plan assumes until Chunhui Zhao decides.

#	Question	Options	Recommendation and why
F2	SIPG fluxes (derivation §6)	(a) total-stress split: the solid block is today's elastic SIPG with λ_u; (b) revision 2.2's ζ-form	**(a).** Consistent for any material jumps (derivation Proposition 6.1; test U2-6), and it reuses today's stiffness, penalty, traction kernel and slip load unchanged (Proposition 6.2; test U2-4).
F3	Velocity coefficients (derivation §9.3)	(a) Γ_{mom}; (b) Γ_{diag}	**(a).** Total momentum advances by the trapezoid rule at no extra cost (Propositions 9.2, 9.3; test U2-14). Its stability is measured by U2-18.
D9	Pass test for the SeisSol comparisons (§7.3)	(a) the TPV pass test Chunhui Zhao accepted on 16 Oct (arrival 0.1 s, final slip 2 %, peak slip rate 10 %), plus misfits that shrink with refinement in both codes and a finest-level gap no larger than SeisSol's own ORDER 3 to 4 spread; (b) only "converge towards a common answer"	**(a).** It is the rule our elastic golds already pass, so a poroelastic result is held to the same standard, and the spread condition stops a coarse SeisSol run from being treated as exact.
D10	Allocation for the 100 m runs	(a) EAR24034 (8862 SU); (b) EAR26024 (72 SU at the last check, expires 30 Nov)	**(a).** The 100 m runs are the largest, and SeisSol's cost there is not known until the probe of §9 (ours is about 8 SU, inferred). EAR26024 has 72 SU left, shared with the other benchmark threads, and expires on 30 Nov; EAR24034 has 8862 SU. The builds, S0, S1 and the 200 m runs stay on EAR26024.
D11	Which SeisSol comparisons	S0 (rebuild check), S1 (bulk pressure pulse, no fault), S2 (`tpv5-poro` rupture); S3 (undrained-equivalent TPV205) and S4 (contrast fault) optional	**S0–S2 now, S3–S4 after S2.** S1 tests the bulk and the starting-pressure load without friction, so a disagreement in S2 can be placed in the bulk or the fault. Our runs leave out damping, as SeisSol has none; one extra run of ours with the usual damping shows its effect.
D12	Bring trunk `4c97e32` into the branch first (§4.4)	(a) merge it as step P0.0, before the hooks; (b) stay at `a4c9dee` and merge later	**(a).** Trunk has since changed the three files the hooks touch; merging first means the hooks are written once, and the LTS refusal is tested against real LTS code. The branch adds only two READMEs, so the merge has no conflicts.

2. What the literature says

The full notes, with DOIs, a summary of each paper and its verification status, are in `/mnt/project-files/literature/poroelastic/LITERATURE_NOTES.md`. Paper PDFs are not stored here; the notes give open-access URLs. Section 11 of the derivation document lists the source of every equation and marks the ones derived there.

2.1 DG and high-order methods for Biot waves

Paper	Form	Stiff drag term	What we take
Biot 1956, J. Acoust. Soc. Am. 28(2) 168; Biot 1962, J. Appl. Phys. 33(4) 1482	The (u, w) equations	—	The strong form (derivation §4.2)
de la Puente, Dumbser, Käser, Igel 2008, Geophysics 73(5) T77	First order, ADER-DG	Locally implicit predictor or splitting	Background
Wolf, Galis, Uphoff, Gabriel, Moczo, Gregor, Bader 2022, JCP 455:110886	First order, 13 unknowns, Godunov flux, LTS (SeisSol)	Element-local space–time predictor	The bulk verification set; how SeisSol steps the drag (derivation §9.5)
Dudley Ward, Lähivaara, Eveson 2017 (JCP 350:690); 2021 (CMFT 21:737)	First order, exact Riemann flux	IMEX Runge–Kutta	Warning: at low frequency and permeability, drag errors can dominate
Shukla, Chan, de Hoop, Jaiswal 2020, JCP 403:109061	First order, energy-stable penalty flux	—	A penalty flux close in spirit to SIPG
Antonietti, Botti, Mazzieri, Nati Poltri 2022, SISC 44(1) B1	**Second order** (u, w)**, SIPG**, penalty on $[\![w \cdot n]\!]$ only	Implicit Newmark	The SIPG pattern (derivation §6.2); F2 changes its coupling term
Antonietti, Botti, Mazzieri 2022, Vietnam J. Math. 50:997	Review, same form (eqs. 31–36, read 20 Oct)	Newmark (1/2, 1/4)	The notation check of derivation §4.2
Morency & Tromp 2008, GJI 175:301	Spectral elements	—	Background

Our explicit setting is what the papers do not cover. Antonietti et al. step implicitly, so the drag never limits them; the first-order codes use element-local implicit predictors, IMEX schemes or splitting. The derivation (§9.2) shows why central difference fails on the drag and gives the fitted coefficient that does not; §9.5 there compares it with SeisSol's predictor.

2.2 Rupture and slip in poroelastic media

Paper	What it shows	Use here
Wolf, Gabriel, Galis, Moczo, Gregor, Bader 2025, GJI 242(2) ggaf184 (open access)	3D ADER-DG Biot rupture on tetrahedra: a branching fault and a fault in a damage zone, compared with Gassmann-equivalent elastic runs. They agree at $\alpha = 0.3$; at $\alpha \gtrsim 0.75$ rupture jumps to the branch only in the poroelastic run.	The fault physics to match (derivation §7); the Gassmann-equivalence idea behind V-F1

Rudnicki & Rice 2006, JGR 111 B10308	On-fault pressure from dynamic slip between dissimilar poroelastic materials	Sign and size check for the contrast fault (V-F3)
Dunham & Rice 2008, JGR 113 B09304	2D slip pulses between dissimilar poroelastic materials	Qualitative check: which side's pressure governs
Torberntsson et al. 2018 (Comput. Geosci. 22); Heimisson, Dunham, Almquist 2019 (JMPS 130)	Poroelastic earthquake cycles and rate-and-state stability	V-C1: code-to-code if they share 2D cycle data
Walker et al. 2023, GJI 235(3) (PyLith)	Quasi-static poroelasticity verified on Terzaghi, Mandel and Cryer	V-Q1–V-Q3: the same three tests

2.3 Benchmarks that exist

Benchmark	What it covers	Status for us
`tpv5-poro` (SeisSol precomputed regression)	3D Biot bulk, planar fault, slip weakening, nucleation by a starting pressure; fault and off-fault receivers	Downloaded; the only 3D rupture reference; caveats in §3.1. S0 and S2.
Gar6more 2D/3D (INRIA; Diaz & Ezziani 2010, CiCP 7)	Analytic Green's functions: whole space, free-surface half space, poro/poro bilayers	V-B4; runs on the Mac (Fortran)
Karpfinger, Müller, Gurevich 2009	Full-space poroelastic Green's functions	V-B4
LOHp (Wolf et al. 2022)	Layer over half space with a viscous fluid; finite-difference reference	Optional
SEAS BP6 (Lambert et al. 2025, JGR)	2D antiplane, quasi-dynamic, diffusion along the fault only; a 3D BP6 was announced at EGU26	V-Q7; checks the effective-stress path, not the Biot bulk

3. The benchmarks

3.1 `tpv5-poro` (SeisSol)

Source: `github.com/SeisSol/precomputed-seissol`, recorded 2026-09-28 at SeisSol `3fc8c627` with the poroelastic order-6 double-precision build. Local copy: `/mnt/project-files/literature/poroelastic/tpv5-poro/`.

Item	Value (from `material.yaml`, `fault.yaml`, `parameters.par`)
Domain and mesh	$x, y \in [-60, 60]$ km, $z \in [-60, 0]$ km, 19 183 tetrahedra, edges near the fault 1.0–1.4 km, polynomial degree 5
Bulk	$\mu = 6.40$ GPa, $\lambda = 5.28$ GPa (drained), $K_s = 20$ GPa, $K_f = 2.5$ GPa, $\phi = 0.4$, $\kappa = 6 \times 10^{-13}$ m^2, $\eta = 10^{-3}$ Pa s, $T = 2$, $\rho_s = 2080$, $\rho_f = 1040$ kg/m^3
Fault	Plane $y = 0$; $\sigma_{yy} = -120$ MPa, $\sigma_{xy} = 70$ MPa, uniform; background $p_0 = 0$; linear slip weakening $\mu_s = 0.677$, $\mu_d = 0.525$, $D_c = 0.4$ m (`FL = 16`, no forced rupture time given)

Nucleation	Starting pressure $p_{\text{init}} = 50\,\text{MPa}\,\exp(-10^{-7}r^2)$ about $(0, -1, -7.5)$ km with zero strain and stress (`PressureInjection`); 45.24 MPa at the nearest fault point; end time 2 s
Time stepping	CFL 0.5, clustered local time stepping (`ClusteredLTS = 2`)
Receivers	4 on the fault ($x = \pm 2$ km, depth 5.5 and 7.5 km), 6 off the fault at depth 7.5 km; off-fault output every 5 ms

> **Caveats found by reading the reference data** (`tpv-faultreceivers.h5`, `tpv-receivers.h5`):
>
> 1. The fault-receiver pressure column `P_f` is exactly 0 at every receiver and time, and the rupture-time column is 0. The fault pressure can be checked only through slip, slip rate and traction, and through off-fault pressure near the fault.
> 2. All four fault receivers lie inside the patch that is overstressed at $t = 0$ (radius 3.17 km on the fault, derivation §7.4), so they start slipping within 0.03–0.07 s; arrival times there carry no information. S2 therefore adds the 16 TPV205 stations, where the rupture front arrives.
> 3. Off-fault receivers record pressure up to 45.3 MPa and solid velocity up to 7.06 m/s. Fault receivers are written every 2.37 ms, off-fault receivers every 5 ms. (All re-read on 20 Oct from the files in `precomputed/double/` of the SeisSol repository; test U5-1 pins them.)
> 4. Elements of 1–1.4 km with $D_c = 0.4$ m are coarse. With no convergence study this is a regression record, not a verified solution, and SeisSol's documentation says poroelastic dynamic rupture is "not tested". S0 only checks that our SeisSol build reproduces it.

Our TPV205 meshes come from the same SeisSol TPV5 `.geo` lineage, with the same ± 60 km box, 60 km depth and a 30 × 15 km fault on $y = 0$ (`tpv205/mesh/tpv2053d_200m.geo`). The 200 m mesh has 1 894 583 tetrahedra and 26 228 fault faces (gold README of job 1075916); the 100 m mesh 5 438 053 and 98 984. Within 2 s no wave reaches the outer boundary and returns, so the absorbing boundary does not affect this benchmark.

3.2 Undrained-equivalent TPV205 (our construction)

Biot constants whose Gassmann undrained moduli and bulk density equal TPV205's ($\rho = 2670$ kg/m³, $c_s = 3464$, $c_p = 6000$ m/s), with the TPV205 deck otherwise unchanged. On a planar fault in a homogeneous medium the volumetric strain from strike slip is odd in y, so the pore-pressure change is odd too, a permeable fault carries $p_f = 0$, and the normal-stress change on the fault vanishes. With a permeability of 10^{-15} m² the rock barely drains in 12 s (diffusion length about 0.5 m), so the rupture must reproduce the elastic TPV205 run. With F2 the solid block is today's elastic operator built from the undrained sample, Kelvin–Voigt layer included (derivation Proposition 6.2), so the damped comparison with the golds needs no special treatment.

Constant	Value	Note

ϕ, ρ_f, K_f	0.1, 1000 kg/m³, 2.2 GPa	water
K_s, ρ_s	100 GPa, 2855.56 kg/m³	Stiff on purpose, to get $\alpha \approx 0.5$; a verification set, not a rock
K_d, λ, μ	47.946, 26.587, 32.038 GPa	Solved from Gassmann $K_u = K_d + \alpha^2 M = 53.403$ GPa; the deck carries λ to 17 digits, so λ_u equals TPV205's λ to 1.2×10^{-16}
α, M, B	0.5205, 20.14 GPa, 0.1963	
κ, η, T	10^{-15} m², 10^{-3} Pa s, 2	Stiff drag ($z \sim 2 \times 10^5$), which tests D2

Two meshes are used. On the gmsh 200 m mesh the run is compared with the Vista p2 gold, job 1075916 (V-F1). The tets on the two sides of each face of that mesh differ, which is what makes the elastic gold's normal stress noisy, so there p_f is small but not zero. On the 1000 m mirror-symmetric mesh (`scripts/make_mirror_meshes.sh`, commit `41879e8`; small enough for the Mac) the symmetry holds in the discrete problem, so p_f must vanish to round-off and the run must equal the elastic run with (λ_u, μ, ρ) on the same mesh up to the tiny drained correction (V-F2). A second variant breaks the symmetry with a permeability and porosity contrast across the fault; its $p_f \neq 0$ is checked for sign against Rudnicki & Rice (2006) (V-F3).

3.3 Derived constants for both media

Computed by `poro_test_refs.py` from derivation (4.4), (4.6) and (8.4); the unit tests compare against these values (§6.2). The time steps use CFL 0.5 with the fast P speed, as an estimate.

Quantity	**`tpv5-poro`**	**TPV205 undrained-equivalent**
α, M, B	0.5227, 6.019 GPa, 0.2811	0.5205, 20.14 GPa, 0.1963
λ_u, K_u	6.924, 11.19 GPa	32.04, 53.40 GPa
Bulk density ρ, fluid mass m	1664, 5200 kg/m³	2670, 20 000 kg/m³
Drag relaxation time τ	2.730 µs	0.0196 µs
Biot frequency f_c	51.0 kHz	7.96 MHz
Fast P (high frequency) / undrained P / drained P	3590.3 / 3442.9 / 3296.3 m/s	**6029.5** / 6000.0 / 5827.2 m/s
S (high / low frequency)	2096.6 / 1961.2 m/s	3496.9 / 3464.0 m/s
Slow P (high frequency)	1056.0 m/s	**979.0** m/s
Hydraulic diffusivity	3.31 m²/s	**0.0190** m²/s
$z = \Delta t/\tau$ at 200 m, p = 2	2.55×10^3 ($\gamma_w = 0.99961$)	2.11×10^5 ($\gamma_w = 0.999995$)

The three bold values replace revision 2.2's 6051 m/s, 1046 m/s and 0.022 m²/s. Those came from `biot.py`, which still had $K_s = 60$ GPa while the table's other entries used 100 GPa. Nothing else in revision 2.2 depended on them. Seismic frequencies lie far below f_c in both media, so the slow wave is diffusive (it spreads about 2.6 m in 2 s in `tpv5-poro`) and the bulk is effectively undrained; the poroelastic effect on rupture comes through p_f.

3.4 SeisSol on Vista

SeisSol fixes the order when it compiles; its `ORDER` is $p+1$, and poroelastic builds accept ORDER 2 to 8 and select the space–time predictor on their own (`cmake/process_users_input.cmake`, SeisSol master, read 20 Oct). Poroelastic builds also need a Fortran compiler and LAPACK (`CMakeLists.txt:297`).

What Vista has (checked 20 Oct). Under `$WORK/seissol`: four builds, all `EQUATIONS=viscoelastic2` at ORDER 4 and 6, double precision, for the GH200 GPUs, from `src/SeisSol` at `a8e7f04`; a second checkout, `src-gpu/SeisSol-master` at `31f40bb`, has no binary. There is no poroelastic build at any order, no ORDER 2 or 3 build, and no PUMGen. The dependencies and a CPU build kit (`kit/build_seissol_vista.sbatch`) are there.

Proposed builds. Four CPU builds (Grace nodes), `EQUATIONS=poroelastic`, ORDER 2, 3, 4 and 6, double precision, from the `31f40bb` checkout into new build folders, so the existing builds are untouched; ORDER 6 is for S0 only. The build job records whether `3fc8c627`, the commit of the published record, is an ancestor of the source it builds. PUMGen builds only on Linux, so it is built on Vista with the `phdf5` module, in a new clone under `$WORK/seissol`. The build and run scripts are committed to this branch after review, and each job is submitted only on Chunhui Zhao's words (§9).

Mesh pipeline. Our gmsh v2.2 meshes, sha256 recorded, are converted by `pumgen -s msh2 mesh.msh` to `mesh.puml.h5`, the command used for the earlier SAFS SeisSol meshes (`safs/.../preprocess/build_and_run_pumgen_linux.sh`). SeisSol's boundary tags 101 / 103 / 105 are the ones our meshes already carry (`tpv205/configs/tpv205_nitsche.toml:24-25`). For S1 a copy of the mesh has the fault tag removed (tag 103 faces become interior), written by a small script that checks the node and element counts are unchanged.

3.5 Permeable and sealed faults, in plain words

The difference is whether pore fluid can flow across the fault plane. On a **permeable** fault fluid passes freely, so the pore pressure is the same on both sides and friction sees one p_f; this is what SeisSol does in `tpv5-poro`. On a **sealed** fault (a clay-rich gouge or a low-permeability core) no fluid crosses, each side keeps its own pressure, and friction needs a rule for which pressure it sees. Slip squeezes the rock on one side and stretches it on the other; on a permeable fault the two pressure changes meet in the middle and, in a uniform medium, cancel, which is why the undrained-equivalent TPV205 has $p_f = 0$. In code, permeable means the fault faces get the same fluid face terms as an interior face (derivation §7.2); sealed means each side gets a no-flow condition $w^{\pm} \cdot n^{\pm} = 0$ instead (derivation §7.5), which arrives in P6.

4. Code layout and the option

4.1 Folders

```
miniapps/seas/multiphysics/
  README.md                       what lives here, one subfolder per physics
  poroelastic/
    poro_config.hpp/.cpp          parse [multiphysics] tables; abort on missing or unknown keys
    poro_material.hpp             PoroMaterialSample: derivation (4.4) per element, and the
                                  undrained sample (lambda_u, mu, rho) for the solid block
    poro_dispersion.hpp           closed-form speeds of (4.6): fast/slow P, S (CFL and tests)
    dg_poro_div_integrator.hpp    B_c of (6.3), fluid penalty (6.5), boundary rows (§6.6),
                                  source loads (6.6)
    poro_source.hpp               PoroSource: Q(x, t), V = int Q dt, p_init        (4.3), (6.6)
    poro_mass.hpp                 PoroMass: (M + dt D Gamma)^{-1}, built on NitscheMass   (8.3)
    poro_nitsche_operator.hpp     owns a NitscheWaveOperator built from the undrained
                                  sample; the Z blocks; the force of (9.2)
    poro_newmark.hpp              PoroNewmarkStepper, FittedGamma            (9.1)-(9.5)
    poro_fault_pressure.hpp       FacePressure (7.2), CouplingTraction (7.3),
                                  FaultPressureHook (7.4)
    poro_cfl.hpp                  PoroMaxDt                                        (9.6)
    poro_receivers.hpp            off-fault receivers: v, w_dot, p, sigma
    poro_qd_operator.hpp          P7: split step                             (10.2)-(10.3)
    poro_diffusion_solver.hpp     P7: SPD solve of (10.3), cached MUMPS + PCG (D8)
    poro_handoff.hpp              P8: state copy across the switch, derivation §10.3
    tests/
      poro_fixtures.hpp           meshes F-A to F-F (§6.1); uses tests/unit/test_macros.hpp
      poro_test_refs.hpp          generated reference values, 17 digits (§6.2)
      refs/poro_test_refs.py      the generator, with refs.json
      test_poro_p0.cpp ... test_poro_p8.cpp     one file per phase, make test-poro-pN
    benchmarks/
      bulk/                       plane waves, Darcy, undrained, energy (V-B1 to V-B5)
      tpv205_undrained/           decks and comparison scripts (V-F1 to V-F4)
      tpv5_poro/                  our decks, SeisSol parameter files, converters (S0 to S4)
      seissol/                    Vista build and run scripts, mesh conversion
      qd_ladder/                  Terzaghi, Mandel, Cryer, dislocation, MMS, BP6 (V-Q1 to V-Q7)
```

The folders already on the branch (`miniapps/seas/poroelastic/benchmark_document` and `benchmark_data`, commit `5b2c77d`) stay where they are, for papers and reference data. With F2 the integrator carries the divergence form B_c, so revision 2.2's `dg_poro_zeta_integrator.hpp` is renamed `dg_poro_div_integrator.hpp`, and F1 adds `poro_source.hpp`. The existing tests paste their own `TEST_ASSERT` and copy `CreateTestMesh24Tet` into each file (`tests/verification/test_nitsche_gates.cpp:47, 229`); the new tests use the shared `tests/unit/test_macros.hpp` and keep their fixtures in one header.

4.2 The option

```
[multiphysics]
poroelastic = false              # default; true switches on the Biot (u, w) bulk

[multiphysics.poroelastic]       # required when poroelastic = true; every key required
porosity          = 0.4
permeability_m2   = 6.0e-13
viscosity_pa_s    = 1.0e-3
tortuosity        = 2.0
rho_fluid         = 1040.0
rho_solid         = 2080.0
bulk_fluid_pa     = 2.5e9
bulk_solid_pa     = 2.0e10
fault_hydraulics  = "permeable"  # "sealed" arrives in P6
background_p_pa   = 0.0          # p_0, uniform; enters sigma_n' only
fluid_penalty     = 1.0          # kappa_w of (6.5); value fixed by test U2-5
drag_gamma        = "fitted"     # "half" exists only for test U2-15
velocity_coupling = "momentum"   # F3: Gamma_mom; "diagonal" = Gamma_diag

[multiphysics.poroelastic.initial_pressure]   # optional; SeisSol 'PressureInjection'
origin_m     = [0.0, -1000.0, -7500.0]        # p_init = magnitude exp(-width r^2),
magnitude_pa = 5.0e7                          #   with zero strain and zero total stress
width_per_m2 = 1.0e-7

[multiphysics.poroelastic.source]             # optional; injection rate Q(x, t) (F1)
shape         = "gaussian"       # or "well": Q_w(t) delta(x - x_w), x_w inside one element
origin_m      = [0.0, -1000.0, -7500.0]
width_per_m2  = 1.0e-7           # gaussian only
rate          = 1.0e-6           # peak Q in 1/s (gaussian) or Q_w in m^3/s (well)
time_function = "smooth_step"    # "step", "ramp" or "smooth_step"; each has a closed-form
rise_time_s   = 0.1              #   time integral, so V(t) is exact
```

When the option is on, the λ, μ in `[material_constant_fallback]` are the drained frame moduli, and the driver aborts if that table's `rho` differs from the bulk density $(1-\phi)\rho_s + \phi\rho_f$ by more than 10^{-12} relative, rather than silently reinterpreting it. A `[multiphysics.poroelastic]` table present while the option is off is ignored with a warning in the driver log. The command line gets `--poroelastic 0|1`, in the style of `--fast-fault`; `1` aborts when the table is missing. The `fluid_penalty` and `velocity_coupling` keys are new in this revision: the first because the derivation leaves κ_w to be measured, the second so that test U2-14 can run both choices of F3.

4.3 Hooks in existing files (all bit-identical when off)

#	File (lines at `5b2c77d`)	Change
H1	`domain/nitsche_wave_operator.hpp:840-909`	Split `Acceleration_` into `Force_` (the right-hand side) and `Minv.Apply` (line 908), so the coupled operator reuses the elastic force with its own mass. The arithmetic order is unchanged (test U0-3).

H2	`domain/nitsche_wave_operator.hpp:993-1099, 488-516`	An optional `FaultPressureHook*`, null by default. Before box 8 it adds the coupling traction $\{\{\alpha M(\nabla \cdot w - V)\}\}n$ to the trial traction (derivation 7.3), and it passes $p_0 + \hat{p}$ to `CloseDOF`, which subtracts it from σ_n before the range check at line 495 (7.4). $\hat{p}$ is recorded in the step record.
H3	`drivers/nitsche_dyn_driver.cpp:679-710, 880-883, 1061`	Parse `[multiphysics]`. When it is on, build `PoroNitscheOperator` and `PoroNewmarkStepper` in place of today's operator and stepper, and write the new outputs to separate `*_poro.dat` files, so the existing station layout stays byte-identical. P0 also replaces the abort at 880-883 (static `P_p` "not wired") with $\sigma_n' = \sigma_n - P_p$; existing decks have $P_p = 0$.
H4	`Makefile`	Targets `test-poro-p0` ... `test-poro-p8` and `check-poro-refs`; `nitsche_dyn_driver` gains the `multiphysics/poroelastic` objects.
H5 (P7)	`solver/seas_operator.hpp:318-335`	In `Mult`, an optional fluid-state hook, null by default: it adds $-Z_{uw}w^*$ to the right side of `domain_->Solve` and passes $p_0 + \hat{p}$ to the stress balance (derivation §10.2).
H6 (P7)	`drivers/seas_driver.cpp:580-590`	When the option is on, the native RK45 loop calls the diffusion half-steps before and after `ode_solver.Step`; a rejected step restores w^n and redoes the first half-step with the smaller Δt. The PETSc TS path aborts with the option on.
H7 (P8)	Switch code on the fully-dynamic-cycle branch	Carries w across the switch (derivation §10.3). Written only after that branch's switch code is brought in, which is Chunhui Zhao's call at that time.

The static P_p of P0 is independent of Biot, and is also what TPV12/13 (hydrostatic pore pressure) on the complex-geometries branch will need. It enters only σ_n', never the bulk.

Refused with the option on, each with an abort and a message (test U0-6): local time stepping, the GPU paths, the PETSc TS quasi-dynamic path, and, in the quasi-dynamic solver, the opt-in dense Green's function (`--qd-green`, on the quasi-dynamic speed branch), because traction then depends on w as well as on slip, which a slip-only G cannot represent. This follows the project rule that linear-only shortcuts refuse models they cannot represent. The u block keeps today's fused and dense stiffness paths; the Z blocks use plain assembled matrices until P6.

4.4 Where the branch starts (D12)

The branch is at `5b2c77d`, whose code equals trunk `a4c9dee`. Trunk is now `4c97e32`, which adds local time stepping and changes the three files the hooks touch (`domain/nitsche_wave_operator.hpp` +98 lines, `domain/nitsche_mass.hpp`, `drivers/nitsche_dyn_driver.cpp`; `git diff --stat a4c9dee 4c97e32`, read 20 Oct). The branch adds only two README files, so the merge has no conflicts. **D12 (recommended): merge trunk `4c97e32` into the branch as step P0.0, before the hooks are written**, so they are written once against today's trunk and the LTS refusal is tested against real LTS code. Otherwise the hooks would be rewritten at the next trunk merge. The merge gets its own review page; the words that approve it are in §9.

5. Implementation steps

Each step is one review and one commit. The code and its tests are written together, built and run on the Mac in the branch's local repository under the run lock, and put on a review page with the diff, the test output and the commit message. After Chunhui Zhao's words approving that step, this thread commits exactly the reviewed tree and message and pushes, and the Mac checkout moves to the pushed commit; the Mac never commits or pushes. A step whose tests fail is not put up for review: the failure is reported with its output and a diagnosis, and a change of approach is asked for, not made.

Step	What it adds	Tests that end it	Where
P0.0	Merge trunk `4c97e32` (if D12)	U0-2 on the merged tree	Mac
P0.1	`poro_config`: the tables of §4.2, `--poroelastic`, abort rules; `multiphysics/README.md`	U0-4	Mac
P0.2	Hooks H1, H2 (null), H3 (parse only), H4; refusals	U0-1, U0-2, U0-3, U0-6	Mac
P0.3	Static P_p in σ'_n (H3)	U0-5, U0-1 again	Mac
P1.1	`PoroMaterialSample`, undrained sample; `poro_test_refs.py`, its header and `check-poro-refs`	U1-1 to U1-4	Mac
P1.2	`poro_dispersion`, `PoroMaxDt`	U1-5, U1-6	Mac
P2.1	`DGPoroDivIntegrator`: element and face blocks, boundary rows, penalty, source loads	U2-1, U2-2	Mac
P2.2	`PoroNitscheOperator`: $\mathbb{K}$, the force of (9.2), the undrained elastic operator inside	U2-3 to U2-8	Mac
P2.3	`PoroSource`	U2-9	Mac
P2.4	`PoroMass`, composed from `NitscheMass` (below)	U2-10, U2-11	Mac
P2.5	`PoroNewmarkStepper`, `FittedGamma`; H3 builds them	U2-12 to U2-19	Mac
P2.6	Bulk decks and analysis scripts (`benchmarks/bulk/`)	V-B1, V-B2, V-B3, V-B5	Mac
P3.1	Gar6more3D and Karpfinger drivers and comparison	V-B4	Mac
P4.1	`FacePressure`, `CouplingTraction`, `FaultPressureHook` (H2 live)	U4-1 to U4-5, U4-8	Mac

P4.2	Driver: sources and starting pressure with the fault, `*_poro.dat` outputs, receivers	U4-6, U4-7, U4-9, U4-10	Mac
P4.3	Undrained-equivalent and contrast decks (`benchmarks/tpv205_undrained/`)	V-F2, V-F3, V-F4 (Mac); V-F1 (Vista)	Mac, Vista
P5.1	SeisSol and PUMGen build scripts; S0	S0	Vista
P5.2	Deck generator (ours and SeisSol's from one source), mesh converter checks, HDF5 reader, comparison script	U5-1 to U5-3	Mac
P5.3	The comparison runs and their report	S1, S2; S3, S4 if chosen	Vista
P6.1	Dense blocks for Z; timing table	U6-1, U6-3	Mac, Vista
P6.2	Sealed fault (derivation §7.5)	U6-2	Mac
P6.3	User-document section	review only	—
P7.1	`PoroDiffusionSolver`: (10.3), MUMPS with PCG between refactorizations	U7-1	Mac
P7.2	`PoroQDOperator`, hooks H5 and H6	U7-2 to U7-6	Mac
P7.3	Quasi-dynamic ladder decks and analysis	V-Q1 to V-Q6	Mac, Vista
P7.4	Prescribed fault pressure in the antiplane solver; BP6	V-Q7	Mac
P8.1	Hand-off H7 (after the fully dynamic switch code is brought in)	U8-1 to U8-3	Mac
P8.2	Undrained cycle and switch runs	V-C1, V-C2	Vista

5.1 What each step contains

P0.2, the force split (H1). `Acceleration_` today computes the right-hand side and applies the mass inverse in one function. It becomes `Force_(u, d, v*) → F` followed by `Minv.Apply(F) → a`, with every operation in the same order. The elastic driver keeps calling the pair, so its output does not change (U0-3); the poroelastic operator calls `Force_` for the u rows and its own mass.

P1.1, the sample. `PoroMaterialSample(pmesh, mat, cfg)` reads the ten Biot inputs at each element centroid, like `NitscheMaterialSample`, and derives $\rho, \rho_f, m, b, \alpha, M, \lambda_u, \tau, c_{\text{fast}}$ once (derivation 4.4, 8.4). `UndrainedSample()` returns a `NitscheMaterialSample` with (λ_u, μ, ρ); the elastic operator, its penalty, traction kernel, slip load and Kelvin–Voigt layer are all built from it (derivation Proposition 6.2).

```
derived constants, per element (derivation 4.4, 8.4)
  K_d = lambda + 2 mu / 3          alpha = 1 - K_d / K_s
  M   = 1 / (phi / K_f + (alpha - phi) / K_s)
  lambda_u = lambda + alpha^2 M    rho = (1 - phi) rho_s + phi rho_f
  m = T rho_f / phi                b = eta / kappa
  tau = (rho m - rho_f^2) / (b rho)
  c_fast = sqrt(largest eigenvalue of Mrho^{-1} [[lambda_u + 2 mu, alpha M], [alpha M, M]])
```

P2.1, the integrator. `DGPoroDivIntegrator(c_per_elem, penalty)` assembles $B_c(a, b)$ of derivation (6.3) for one coefficient field c. It is used three times: $c = \alpha M$ for Z_{wu} (and its transpose Z_{uw}), and $c = M$ with the penalty σ_w for Z_{ww}. The face block follows the pseudocode of derivation §2.3, with the side signs and the (c_0, c_1, c_2) pattern of the elastic `AssembleFaceBlock`. The boundary rows are those of derivation §6.6, and `AssembleSourceRHS` assembles F_V and F_{init} of (6.6) with the same loop.

P2.2, the operator. `PoroNitscheOperator` owns a `NitscheWaveOperator` built from the undrained sample and three assembled `HypreParMatrix` blocks. Its force is

```
F_u = Force_(u, d, v*_u)  +  F_V,u(t)  -  Z_uw w              existing force, then (6.3),
(6.6)
F_w = F_V,w(t) + F_init  -  Z_wu u  -  Z_ww w  -  b M_e v*_w   (6.3), (6.6), drag of (9.2)
```

P2.4, the mass. `PoroMass` applies (8.3) element by element, using `NitscheMass::ElementMassInverse(e)` (`domain/nitsche_mass.hpp:191`) for M_e^{-1}. It writes the 2×2 inverse in Schur form,

$$W_{uu} = \frac{1}{\rho - \rho_f^2/m'}, \qquad W_{uw} = -\frac{\rho_f}{m'} W_{uu}, \qquad W_{ww} = \frac{1}{m'} + \left(\frac{\rho_f}{m'}\right)^2 W_{uu}, \qquad m' = m + \Delta t\, \gamma_w b,$$

so that with $\rho_f = 0$ the u rows reduce to today's $(1/\rho)\, M_e^{-1}$ bit for bit (U2-11). In elements with an absorbing face, the dense block is inverted by the same Schur complement around `NitscheMass`'s own block $(\rho M_e \otimes I_3 + \frac{\Delta t}{2} C_e)^{-1}$, which keeps the same property there.

P2.5, the stepper. `PoroNewmarkStepper::Step(x, v, a, t, dt)` is derivation §2.2. `FittedGamma(z)` returns $\frac{1}{2} + z/12 - z^3/720 + z^5/30240$ for $z < 10^{-2}$ and $1/(1 - e^{-z}) - 1/z$ otherwise (derivation 9.4); γ_w is computed per element once per Δt. The u-row arithmetic mirrors MFEM's `NewmarkSolver::Step` operation by operation (`linalg/ode.cpp:1271-1304`; the elastic driver's `CentralDifferenceSolver` is that step with $\beta = 0$, $\gamma = \frac{1}{2}$), so that U2-15 can hold bit for bit.

P4.1, the fault. `FacePressure(x, V)` evaluates $\hat{p} = \{\{p_h\}\} + \sigma_w [\![w \cdot n]\!]$ (7.2) at the fault quadrature points in the loop of `ComputeTractionAtQuadPoints` and projects it to fault DOFs as the traction is projected. `CouplingTraction(w, V)` returns $\{\{\alpha M(\nabla \cdot w - V)\}\} n$ per fault DOF (7.3). `FaultPressureHook` holds both and is what H2 calls; with the hook null, box 8 runs today's code.

5.2 One dynamic step

The step is derivation §2.2; for reading convenience it is repeated here, with the test that guards each line.

```
PoroNewmarkStepper::Step(x = [u; w], v, a, t, dt)
  1  x^{n+1} = x^n + dt v^n + dt^2/2 a^n                          U2-15, U2-17
  2  v* = v^n + dt (I - Gamma) a^n                                 U2-13, U2-14
  2b V^{n+1} from PoroSource (exact, or trapezoid)                 U2-9
  3  op.AdvanceFault(x^{n+1}, t_{n+1}, v*)       boxes 3-11, through H2:
       6  T_h = today's traction kernel with lambda_u              U2-4, U4-4
       6b T = T_h + avg(alpha M (div w - V)) n                     U4-2
       6c p_hat = FacePressure(x^{n+1}, V^{n+1})                   U4-1
       8  sigma_n' = -(T + S).n - (p_0 + p_hat)                    U4-3, U0-5
       9  closure, gamma = sigma_u dt/2 (unchanged)                U4-4, U4-5
  4  F = PoroNitscheOperator::Force(x^{n+1}, d^{n+1}, v*, t_{n+1})  U2-3 to U2-8
  5  a^{n+1} = PoroMass(Gamma).Apply(F)                            U2-10, U2-11
  6  v^{n+1} = v* + dt Gamma a^{n+1}                               U2-14, U2-16
```

The quasi-dynamic split step (P7) and the fully dynamic hand-off (P8) are as in derivation §10 and plan revision 2.2 §5.3–5.4; their tests are in §6.4.

6. Unit tests

Every new function gets a test before it is used by another, and every test has an expected value fixed before the code runs. The tests follow today's conventions: plain C++ with `TEST_ASSERT` and `TEST_NEAR`, a `PASSED` or `FAILED` line per check, one binary per phase run by its Makefile target with `$(MFEM_MPIEXEC) $(MFEM_MPIEXEC_NP)` at 1, 2 and, where ranks matter, 4 ranks, as `test-nitsche-hook-spy` does (`Makefile:3279-3284`). Abort checks run the binary with a flag and grep its message, as that target does for `--abort-reject`.

6.1 Fixtures

Fixture	What it is	Used by
F-A	Two tetrahedra sharing one face, fixed non-symmetric vertex coordinates, $p = 1$–3	element and face matrices; element stepping
F-B	The 24-tet TPV102 box of `CreateTestMesh24Tet` (`test_nitsche_gates.cpp:229`), fault attribute on $Y = 0$, mirror-symmetric under $Y \to -Y$ (checked at `:334`), $p = 1$–3, 1, 2 and 4 ranks	assembly, symmetry, operator identities, fault
F-C	F-B with medium A (`tpv5-poro`) for $x < 0$ and medium B (TPV205-equivalent) for $x > 0$; the jump is on an interior plane of faces	patch tests across a material jump
F-D	Periodic cube of side 1 km, 6 tets per hex from MFEM's Cartesian generator, 4, 8 and 16 hexes per side, made periodic with `Mesh::MakePeriodic`	plane waves (V-B1); energy
F-E	TPV205 meshes: gmsh 1000 m and the 1000 m mirror mesh (Mac); gmsh 200 m and 100 m (Vista); copies with the fault tag 103 removed	off-path identity; fault tests; SeisSol runs
F-F	Quasi-dynamic shapes: a 1D column, a thin slab and a sphere, all tetrahedral	Terzaghi, Mandel, Cryer

Whether our integrators accept MFEM's periodic meshes, which carry a discontinuous nodal field, is checked at the start of P2.6 (not yet verified). If they do not, F-D uses the exact solution's displacement and pressure as boundary data on all faces instead. The mirror mesh

is used here only as a test fixture: Chunhui Zhao dropped mirror meshes for production runs on 17 Oct because SAFS geometries cannot rely on them, but its exact symmetry gives an answer known to round-off, which no other fixture gives.

6.2 Reference values

`poro_test_refs.py` (NumPy and SciPy, independent of the C++ code) computes every expected value; `--json` writes `refs.json` and, from P1.1, `--header` writes `poro_test_refs.hpp` with 17 significant digits for the C++ tests. `make check-poro-refs` regenerates the header and fails on any difference, so the committed values cannot drift from the script. Its functions:

Function	What it computes
`derive`	The derived constants of derivation (4.4), (4.6) and (8.4), and the high- and low-frequency speeds, from the ten inputs
`tpv205_equivalent`	The drained K_d (Brent's method) whose Gassmann K_u matches TPV205; then `derive`
`dispersion`, `modes_real_k`	Biot dispersion: complex wavenumber at real frequency, and complex frequency at real wavenumber (a 4 × 4 companion eigenproblem), for the plane-wave tests
`fitted_gamma`, `gamma_table`	$\gamma_w(z)$ with its series below $z = 10^{-2}$, and z, γ_w for each mesh and order
`tpv5_start`	p_{init}, σ_n' and strength at the hypocentre, the critical pressure, the overstressed radius, and the instant-injection value
`patch_fields`	The exact piecewise-linear static solution across a material jump (PT-1, PT-2)

Values the tests use (from `refs.json`, 20 Oct; quoted to 7 digits here, used at full precision):

Quantity	Value	Test
`tpv5-poro` α, M, λ_u, τ	0.5226667, 6.019262 GPa, 6.924345 GPa, 2.730000 µs	U1-1
TPV205-equivalent λ_d, λ_u	26.58745 GPa, 32.04376 GPa (TPV205's λ to 1.2×10^{-16})	U1-1
$\gamma_w(1)$, $\gamma_w(10)$, $\gamma_w(2600)$	0.5819767068693, 0.9000454019910, 0.9996153846154	U2-12
Series against closed form at $z = 10^{-2}$	1.3×10^{-14} apart	U2-12
Plane waves, 1 km, `tpv5-poro`	P 3442.902 m/s (decay 5.062×10^{-5}/s), S 1961.161 m/s (2.961×10^{-5} /s), diffusive mode decay 1.307×10^{-4}/s, drag mode 3.663×10^{5}/s	V-B1
Plane waves, 1 km, $\kappa = 10^{-7}$ m²	fast P 3588.811 m/s (0.09586/s), slow P 1044.282 m/s (1.003/s), S 2093.324 m/s (0.1345/s)	V-B1
PT-1 slopes (A, B)	u': -5.530973×10^{-5}, -1.102977×10^{-5}; w': 2.890855×10^{-5}, 5.741416×10^{-6}	U2-6
PT-2 slopes ($V = 10^{-4}$)	w': 1.289086×10^{-4}, 1.057414×10^{-4} (u' as PT-1)	U2-7

`tpv5-poro` start	p_{init} 45.24187 MPa, σ_n' 74.75813 MPa, strength 50.61125 MPa, p_{crit} 16.60266 MPa, radius 3166.165 m; instant injection 98.40455 MPa	U4-6, U4-10

6.3 Tolerance classes

Class	Meaning	Typical bound
B	Bit for bit: `memcmp` of the doubles, or byte-identical output files	exact
R	Round-off: the two sides are the same number computed in a different order, relative to the size of the largest term summed	10^{-14} for one evaluation, 10^{-12} for an assembled vector, 10^{-10} after many steps
C	Convergence: an observed rate from three or more refinements in a stated window	spatial L^2 rate $\geq p + 1 - 0.3$; temporal error ratios in $[3.5, 4.5]$ per halving
D	Discretization level: a comparison with a reference that has its own error, with a stated pass test	per test

A bound is fixed in the test before it first runs. If a test fails, the bound is not changed to make it pass: the failure is reported with its numbers, and a change of bound needs Chunhui Zhao's approval with the reason recorded in the test.

6.4 The tests

P0 (`test_poro_p0.cpp`, `make test-poro-p0`):

ID	What it checks	Expected	Tol.
U0-1	The off path. TPV205 gmsh 1000 m, $p = 2$, the gold deck (Kelvin–Voigt 0.1 and Prakash–Clifton), 200 steps, and the TPV102 inline box: option absent, `poroelastic = false`, and the base build	station, fault and log data files identical	B
U0-2	Every `test-nitsche-*` target and `make test`	same pass counts as at the base commit	B
U0-3	H1: `Force_` then `Minv.Apply` against the old `Acceleration_`, on F-B, $p = 1$–3, with and without damping and absorbing faces, 1 and 2 ranks	identical acceleration	B
U0-4	The parser: each required key missing, misspelt or of the wrong type; an unknown key; a `rho` mismatch; `--poroelastic 1` without the table	abort, naming the key; table present with the option off gives a warning	B
U0-5	Static P_p: F-B with $P_p = 10$ MPa and $\sigma_{yy} = -120$ MPa against $P_p = 0$ and $\sigma_{yy} = -110$ MPa, 500 steps	same fault and station data; bulk identical	R (10^{-10}); bulk B
U0-6	Refusals with the option on: LTS (with D12), GPU, PETSc TS, `--qd-green`	abort, each with its message	B

P1 (`test_poro_p1.cpp`):

ID	What it checks	Expected	Tol.
U1-1	Derived constants for `tpv5-poro`, the TPV205-equivalent set and the V-F3 contrast set	`poro_test_refs.hpp`	R (10^{-13})
U1-2	Identities: $K_u = K_d + \alpha^2 M$; $B = \alpha M / K_u$; $\lambda_u - \lambda = \alpha^2 M$; Gassmann's K_u; $\tau = (m - \rho_f^2/\rho)/b$	each side equal	R (10^{-14})
U1-3	Limits: $K_s \to \infty$ gives $\alpha = 1$, $M = K_f/\phi$; the TPV205-equivalent λ_u equals $\rho(c_p^2 - 2c_s^2)$ of TPV205	as stated	R (10^{-15})
U1-4	Sampling on F-C: each element carries its side's constants; the undrained sample's (λ_u, μ, ρ) equal the sample's own	identical	B
U1-5	Closed-form speeds: fast and slow P (high frequency), S (high and low frequency), undrained and drained P	`poro_test_refs.hpp`	R (10^{-13})
U1-6	`PoroMaxDt` on F-B and the 1000 m mesh against (9.6) evaluated in the test; on 1, 2 and 4 ranks	identical, and the same on every rank count	B

P2 (`test_poro_p2.cpp`):

ID	What it checks	Expected	Tol.
U2-1	Volume block of B_c on F-A, $p = 1$–3: symmetric; equal to a separate quadrature loop written in the test; zero on rigid motions and divergence-free fields; for linear fields $a = Ax$, $b = Bx$ equal to $c\,\mathrm{tr}A\,\mathrm{tr}B\,\lvert K\rvert$	as stated	R (10^{-14})
U2-2	Face block on F-A: zero for any pair of continuous fields; unchanged when the two elements swap sides; for discontinuous linear fields equal to the closed-form face integral; σ_w equal to (6.5) by hand	as stated	R (10^{-14})
U2-3	Assembled $\mathbb{K}$ on F-B and F-C, $p = 1$–3: $\lvert x^T \mathbb{K} y - y^T \mathbb{K} x\rvert \le 10^{-14} \lvert x\rvert\lvert\mathbb{K}\rvert\lvert y\rvert$ for random x, y; $\mathbb{K}x$ the same on 1, 2 and 4 ranks	symmetric; rank-independent	R
U2-4	Locked fluid (derivation Proposition 6.2): with $w = 0$, the u rows of the force against today's elastic operator built from (λ_u, μ, ρ), slip, Dirichlet and damping on	identical	B
U2-5	Coercivity on F-B and F-C, $p = 1$–3, smallest eigenvalues by LOBPCG: no eigenvalue of $\mathbb{K}$ below $-10^{-10}\lambda_{\max}$ at half the chosen κ_w (a margin of 2); the divergence-free w modes appear as zero eigenvalues	no negative eigenvalue	R
U2-6	Patch test PT-1 on F-C (derivation Proposition 6.1): $\sigma_{xx} = -1$ MPa, drained, $u(0) = 1$ mm, exact u on the boundary; residual of the interpolated exact solution. The revision 2.2 form would leave a jump $\lVert(\alpha u + w)\cdot n\rVert = 2.13 \times 10^{-6}$ m that does not shrink with the mesh	residual zero	R (10^{-12})

U2-7	PT-2: PT-1 with an injected volume $V = 10^{-4}$; and F_{init} for a quadratic p_{init} against $-\int \nabla p_{\text{init}} \cdot q$ plus its boundary term	residual zero; loads equal	R (10^{-12})
U2-8	Consistency on F-B: for (u, w) polynomials of degree p on the whole box, $\mathbb{K}x_I$ against the strong-form residual of (4.1)–(4.3) integrated against each test function, with the matching boundary terms	equal	R (10^{-12})
U2-9	`PoroSource`: $V(t)$ exact for each time function; trapezoid V second order (ratios in $[3.5, 4.5]$); a well's load equals $MV_w\zeta(v, q)(x_w)$ with no face terms; a well on a face or vertex aborts	as stated	R (10^{-15}), C
U2-10	`PoroMass::Apply` against an independently assembled $\mathbb{M} + \Delta t\,\mathbb{D}\Gamma$, for $z = 10^{-2}, 1, 2600, 2 \times 10^5$, with and without absorbing faces, $p = 1$–3	inverse	R (10^{-12})
U2-11	With $\rho_f = 0$: the u rows of `PoroMass` against `NitscheMass`, in plain and absorbing elements	identical	B
U2-12	`FittedGamma`: the three values of §6.2; series and closed form within 5×10^{-14} at the switch; increasing and inside $[\frac{1}{2}, 1)$ on $z \in [10^{-8}, 10^8]$; $R(z) = e^{-z}$ there	as stated	R (10^{-15} absolute)
U2-13	Element relaxation on F-A with uniform fields and a constant force (derivation §9.3), $z = 10^{-2}, 1, 100, 2600$: the relative velocity's error falls as e^{-nz} over 50 steps; from rest, the first velocity is $\gamma_w z/(1 + \gamma_w z)\, v_\infty$ (derivation §9.4); with `drag_gamma = "half"` at $z = 2600$ the error alternates in sign	as stated	R (10^{-13})
U2-14	Momentum (derivation Proposition 9.2), same element, time-varying force: with Γ_{mom} the total momentum changes by $\frac{\Delta t}{2}(f^n + f^{n+1})$; with Γ_{diag} the extra term is $\rho_f \Delta t(\gamma_w - \frac{1}{2})(a_w^{n+1} - a_w^n)$; the fluid velocity and acceleration are the same for both	as stated; fluid identical	R (10^{-13}); B
U2-15	With $\alpha = 0$, $\rho_f = 0$ and `drag_gamma = "half"`, on F-B with slip, 1000 steps: the u block against today's elastic driver step	identical	B
U2-16	Energy on F-B with free surfaces on every face and no loads, 1 and 2 ranks: (a) without drag ($b = 0$, so $\Gamma = \frac{1}{2}I$) the modified energy $E_h - \frac{\Delta t^2}{8} a^T \mathbb{M} a$ of the elastic test T-10 (`test_nitsche_p3.cpp:13`) over 10^4 steps; (b) with drag, the same energy from step to step	(a) constant; (b) never increases	R (10^{-10})
U2-17	Temporal order on F-D, resolved drag ($\kappa = 10^{-7}$ m²): errors against a run at $\Delta t/16$ as Δt halves	second order	C

U2-18	Stability by bisection on blow-up over 1000 steps, as T-09(c) (`test_nitsche_p3.cpp:5-8`), on a 3 × 3 × 2 box of F-B, $p = 1$–3: (a) without drag, the limit against $2/\omega_{\max}$ by power iteration on $\mathbb{M}^{-1}\mathbb{K}$; (b) with the `tpv5-poro` drag, for both `velocity_coupling` choices, the limit against (a); (c) `PoroMaxDt` at CFL 0.5 against the limits	(a) within 1 %; (b) not below (a) by more than 5 %; (c) below both	D
U2-19	The whole step on F-B against 1, 2 and 4 ranks, 200 steps	same fields	R (10^{-12})

P4 (`test_poro_p4.cpp`):

ID	What it checks	Expected	Tol.
U4-1	`FacePressure` on F-B: for continuous (u, w), $\hat{p}$ equals p_h at the fault points; swapping the fault's sides leaves it unchanged; a set jump $[\![w \cdot n]\!]$ raises it by $\sigma_w[\![w \cdot n]\!]$	as stated	R (10^{-14})
U4-2	Total traction (7.3) for a uniform $\nabla \cdot w$ and V against $\alpha M(\nabla \cdot w - V)n$ plus the elastic part by hand	equal	R (10^{-14})
U4-3	σ_n' (7.4): a prescribed uniform p on both sides gives $\sigma_n - p_0 - p$; a rise in p lowers the closure's strength	as stated	R (10^{-14})
U4-4	The closure is unchanged (Proposition 7.1): `ClosureGamma` with λ_u against today's; a spy closure, as in `test_nitsche_hook_spy.cpp`, sees σ_n' of (7.4) at every call	identical	B
U4-5	Slip carries nothing into the coupling block: the w rows of the force with and without a tangential slip field	identical	B
U4-6	`tpv5-poro` start at $t = 0$ on the 1000 m mesh: $\sigma_n' = 120\,\mathrm{MPa} - p_{\mathrm{init}}$ at every fault DOF; the overstressed patch radius	§6.2 values; radius within one element	R; D
U4-7	A uniform p_0 leaves the bulk at rest and enters only σ_n'	bulk zero; σ_n' shifted by p_0	B; R
U4-8	Null hook: with the option on and the hook disabled, the elastic fault path against U2-15's run	identical	B
U4-9	Outputs: `*_poro.dat` columns, units and sign conventions; existing station files unchanged in layout	header and values as specified	B
U4-10	Instant injection ($Q = (p_{\mathrm{inj}}/M)\delta(t)$) at a fault point in the first step: σ_n' drops by $(1-\alpha)p_{\mathrm{inj}}$, not p_{inj} (derivation §7.4)	as stated	R (10^{-12})

P5 (`test_poro_p5.py`, Python, run by `make test-poro-p5`):

ID	What it checks	Expected	Tol.
U5-1	The SeisSol HDF5 reader on the published `tpv5-poro` record: `P_f` and `RT` are 0 everywhere, off-fault p peaks at 45.29 MPa and $\|v\|$ at 7.06 m/s, fault output every 2.369 ms (847 rows), off-fault every 5 ms (401 rows)	as read on 20 Oct	B
U5-2	The comparison script on synthetic traces with known shifts, scalings and noise: it reports the planted arrival shift, slip and peak errors	planted values	R
U5-3	The deck generator: our TOML and SeisSol's `parameters.par`, `material.yaml` and `fault.yaml` from one source carry the same numbers; for `tpv5-poro` it reproduces the published files	same values; published files reproduced	B

P6: U6-1, the dense Z blocks against the assembled ones (R, 10^{-12}); U6-2, the sealed fault: no flux across it, each side's pressure as derivation §7.5, and a permeable run unchanged when sealing is off (R, B); U6-3, a timing table on the Mac and Vista, recorded, with no pass bound.

P7 (`test_poro_p7.cpp`):

ID	What it checks	Expected	Tol.
U7-1	Matrix (10.3) is symmetric positive definite on F-B; MUMPS against PCG with the stale factorization as preconditioner	SPD; same solution	R (10^{-10})
U7-2	Undrained limit ($\kappa \to 10^{-20}$ m²): one quasi-dynamic solve against today's elastic quasi-dynamic solve with λ_u	equal	R (10^{-8})
U7-3	Drained limit (large κ, long Δt): the solve against today's with the drained λ	equal	R (10^{-8})
U7-4	A rejected RK45 step restores w^n and redoes the first half-step	state after retry identical to a run that took the smaller step directly	B
U7-5	BP5 quasi-dynamic with the option off, against the base build	identical	B
U7-6	Splitting order on F-B with a manufactured source: errors as Δt halves	second order	C

P8: U8-1, a hand-off and back with no time elapsed returns the state bit for bit (B); U8-2, energy and w continuous across both switch directions (R, 10^{-10}); U8-3, the fully dynamic cycle with the option off against its base build (B).

6.5 Which test guards which equation

Derivation	What	Unit tests	Verification (§7)
(4.1)–(4.3)	Strong form, sources	U2-8, U2-9	V-B1, V-B2, V-B4
(4.4), (8.4)	Derived constants	U1-1 to U1-4	all
(4.5), Prop. 6.4	Energy	U2-16	V-B5

(4.6), (9.6)	Speeds, time step	U1-5, U1-6, U2-18	V-B1
(6.2)–(6.3)	SIPG form, B_c	U2-1 to U2-3, U2-8	V-B1
(6.4)–(6.5)	Penalties	U2-2, U2-5, U2-18	—
Prop. 6.1	Consistency across material jumps (F2)	U2-6, U2-7	V-F3
Prop. 6.2	Locked fluid is today's elastic operator	U2-4, U2-15	V-B3, V-F1, V-F2
(6.6), §6.6	Source loads, boundary rows	U2-7, U2-9	V-Q1 to V-Q3, S1
(7.1)	Slip load	U2-4, U4-5	V-F1
(7.2)	Fault pressure	U4-1	V-F2, V-F4, S2
(7.3)–(7.4)	Total traction, σ'_n	U4-2, U4-3, U4-6, U4-7, U4-10	V-F1 to V-F3, S2
Prop. 7.1	Closure unchanged	U4-4, U4-8	V-F1
(8.2)–(8.3)	Mass and drag	U2-10, U2-11	V-B1, V-B2
(9.1)–(9.4)	Step, γ_w	U2-12, U2-13, U2-15, U2-17	V-B1, V-B2
(9.5), Props. 9.2–9.3	Γ (F3)	U2-14, U2-18	V-B5
(10.1)–(10.3)	Quasi-dynamic split step	U7-1 to U7-6	V-Q1 to V-Q7
§10.3	The switch	U8-1 to U8-3	V-C1, V-C2

7. Verification

The unit tests check each function against an exact value. The verification tests check whole runs against known answers (§7.1), against our own elastic results through limits the physics fixes (§7.2), and against SeisSol run by us on the same meshes at the same orders (§7.3). The quasi-dynamic and cycle ladders are in §7.4. Each test names its pass test before it runs; the comparison scripts and decks are committed with the step that adds them.

7.1 Bulk, no fault (P2, P3)

ID	Setup	Compared with	Pass test	Where
V-B1	Plane waves on F-D (1 km wavelength), $h = 250, 125, 62.5$ m, $p = 1$–3, starting from an exact mode; two media: a resolved drag (`tpv5-poro` with $\kappa = 10^{-7}$ m², $\tau = 0.455$ s), with fast P, slow P and S; and `tpv5-poro` itself (stiff), with fast P and S	The Biot modes of `poro_test_refs.py` (§6.2)	L^2 error of u, w and p at the end time converges at a rate $\geq p + 1 - 0.3$; at the finest level the measured phase speed within 10^{-3} and the decay rate within 1 % of the reference	Mac

V-B2	Darcy limit: F-B in `tpv5-poro` with $p_{\text{init}} = Gx$ and nothing else	Darcy's law: relative velocity $-(\kappa/\eta)G$	From the second step on, inside the region the boundary has not yet reached (more than n face layers from it at step n), equal to 10^{-10} relative; with `drag_gamma = "half"` the error alternates in sign (documents D2)	Mac
V-B3	Undrained limit: plane P and S waves in the TPV205-equivalent medium with $\kappa = 10^{-20}$ m²	Our elastic run with (λ_u, μ, ρ) on the same mesh and step	Fields within 10^{-6} relative (inferred bound; the fluid moves about 10^{-13} m)	Mac
V-B4	Point sources in a whole space, a free-surface half space and a poro/poro bilayer: an explosive or fluid-injection source, whichever both the reference and our source support (checked in P3.1)	Gar6more3D (Fortran, runs on the Mac); Karpfinger et al. (2009) full-space Green's functions	Relative L^2 misfit of solid velocity and pressure at receivers falls with refinement at the V-B1 rate. The receiver near the free surface is reported separately (Wolf et al. 2022 found that slow wave needs fine meshes)	Mac
V-B5	Energy with resolved drag on F-B with free surfaces	Derivation Proposition 6.4: $\dot{E}_h = -\int b\lvert\dot{w}\rvert^2$	$\lvert E(t) + \int_0^t D - E(0)\rvert / E(0)$ falls at second order as Δt halves, for both choices of F3	Mac

7.2 Fault (P4)

ID	Setup	Compared with	Pass test	Where
V-F1	Undrained-equivalent TPV205 (§3.2), 12 s, Kelvin–Voigt 0.1 and Prakash–Clifton as the golds: (a) gmsh 1000 m, $p = 2$; (b) gmsh 200 m, $p = 2$	(a) our elastic run with TPV205's moduli on the same mesh; (b) the Vista $p = 2$ gold, job 1075916	The TPV pass test of 16 Oct (arrival within 0.1 s at every station, final slip within 2 % where the reference slips more than 0.1 m, peak slip rate within 10 % at 14 or more of the 16 stations); the distribution of $\lvert\hat{p}\rvert$ on the fault is reported	(a) Mac; (b) Vista
V-F2	The same on the 1000 m mirror mesh, $p = 1$ and 2, 12 s	Our elastic run with TPV205's moduli on the same mirror mesh	$\max \lvert\hat{p}\rvert$ over time and fault DOFs below 10^{-6} of the largest bulk pressure change (round-off; symmetry makes it zero); slip, slip rate and traction within 10^{-4} relative (the drained correction, inferred 10^{-6}–10^{-5})	Mac
V-F3	Contrast fault: the same undrained moduli and density on both sides, so the elastic problem stays symmetric, but permeability differing by a factor of 100 and different Skempton coefficients (the V-F3 set of U1-1); 1000 m, $p = 1$–3	Rudnicki & Rice (2006): the on-fault pressure change at a passing rupture front between dissimilar poroelastic media (their formula checked against the paper before use)	Sign of $\hat{p}$ as theirs for the sliding direction; size within a factor that shrinks with refinement	Mac
V-F4	A locked fault is invisible: the `tpv5-poro` start on the 1000 m mesh with a static friction too high to slip, 1 s, undamped; and the same on the copy without the fault tag	The two runs	Bulk fields equal to 10^{-10} relative; $\hat{p}$ on the fault equal to the face pressure of the run without the fault, evaluated on the same faces (inferred: with no slip the fault faces carry the interior-face terms)	Mac

7.3 Comparison with SeisSol (P5)

SeisSol is the only other code with 3D poroelastic dynamic rupture on tetrahedra, but its published `tpv5-poro` record is a single coarse regression run (§3.1). So we run SeisSol ourselves, on our meshes, at the same orders, and compare the two codes at every resolution. SeisSol's `ORDER` is $p + 1$. SeisSol has no Kelvin–Voigt damping or Prakash–Clifton regularization, so our comparison runs leave both out and solve the same equations; one extra run of ours with the usual damping shows its effect. SeisSol keeps its clustered local time stepping (`ClusteredLTS = 2`, as published); our runs use global steps.

ID	What runs	What it shows	Pass test
S0	Our SeisSol build (`31f40bb`, poroelastic, ORDER 6, double, CPU) on the published case: its own mesh `tpv5-20k.h5` and `parameters.par`, 2 s	That our build reproduces the published record (SeisSol `3fc8c627`)	SeisSol's own regression tolerances in `tpv-data.json` (10^{-12} in double). A larger difference is reported with its size, and whether `3fc8c627` is an ancestor of `31f40bb`, before anything else runs
S1	No fault: the 200 m mesh copy without tag 103, the `tpv5-poro` bulk and starting pressure, 2 s; both codes, $p =$ 1–3; the six published off-fault receivers plus a line of receivers through the pulse	The bulk and the starting-pressure load without friction. The relative fluid velocity reaches its Darcy value within the first step, and the total momentum kept by the solid sends out weak waves (velocities of order 10^{-6} m/s, inferred); F3 matters exactly here	Relative L^2 difference of v, p and $\dot{w}$ at the receivers falls with refinement in both codes; at the finest level it is no larger than SeisSol's own $p = 2$ to $p = 3$ spread
S2	`tpv5-poro` on our TPV205 meshes: 1000 m ($p =$ 1–3; also measures SeisSol's cost), 200 m ($p =$ 1–3), 100 m ($p =$ 1–2); the four published fault receivers, the six off-fault receivers and the 16 TPV205 fault stations, where the rupture front arrives	The whole problem: bulk, fault pressure, effective normal stress, friction	D9 (a): at the 16 stations the TPV pass test between the codes at each level; the misfit falls with refinement in both codes; at the finest level our–SeisSol gap no larger than SeisSol's own $p = 2$ to $p = 3$ spread. At the four inner receivers slip rate and slip are compared but carry no arrival information (§3.1)
S3 (opt.)	Undrained-equivalent TPV205 in SeisSol, 200 m, $p = 2$, undamped; with our undamped V-F1	SeisSol's poroelastic rupture against our poroelastic and elastic results on a problem whose answer is known	TPV pass test against our undamped run
S4 (opt.)	The V-F3 contrast fault in SeisSol	Code-to-code on $\hat{p} \neq 0$	Same sign of p_f; gap falls with refinement

The comparison report states the caveats of §3.1 and plots, for every station, both codes at every level, their differences, and the published record where it applies. The three-way deck generator (U5-3) writes our TOML and SeisSol's `parameters.par`, `material.yaml` and `fault.yaml` from one source, so the two codes cannot be set up differently by accident.

7.4 Quasi-dynamic and cycles (P7, P8)

No 3D poroelastic cycle benchmark exists, and SEAS BP6 is 2D. So P7 is verified by a ladder in which every rung has a known answer; only V-Q7 needs outside data.

ID	Test	Pass test
V-Q1–V-Q3	Terzaghi (1D column), Mandel (thin 3D slab) and Cryer (sphere), the PyLith and GEOS set (Walker et al. 2023). Mandel and Cryer show a pressure overshoot that exists only with two-way coupling. They need no-flow rows and rollers; whether the elastic code supports rollers is checked at the start of P7 (not yet verified)	Pressure and displacement histories converge to the closed forms at the spatial rate of V-B1 and at second order in time
V-Q4	A sudden slip patch on a fault in a poroelastic body: just after, our elastic quasi-static solution with (λ_u, μ); long after, with (λ, μ); between, the 2D suddenly introduced dislocation (Rice & Cleary 1976; Rudnicki 1987; both checked before use) with a 3D model uniform along strike	Both limits to discretization error; the 2D comparison converges
V-Q5	A fluid injection rate in a whole space (Rudnicki 1986, point fluid source; checked before use): the general Q of F1 in the quasi-dynamic solver	Pressure and displacement histories converge to the closed form
V-Q6	A manufactured solution with rate-and-state on the fault (the SEAS verification method of Erickson and Dunham; reference to be verified)	Space order $p+1$, time order 2; measures the splitting error of D7
V-Q7	BP6 (Lambert et al. 2025) in our antiplane solver with a prescribed fault pressure, then the same in 3D uniform along strike	Station series within the spread of the BP6 codes; 3D reproduces 2D
V-C1	The undrained cycle: Biot constants whose undrained moduli and density equal BP5's, very low permeability, on the fully dynamic cycle	Event times, slip and station traces as our elastic BP5-FD run, to discretization and solver tolerance
V-C2	Switch checks: energy and w continuous across both directions; an event run fully dynamic from a quasi-dynamic state against the stand-alone dynamic solver from the same state	R (10^{-10}) and agreement to discretization level

8. Risks and open questions

1. **Coupled fitted coefficient.** γ_w is exact for the relaxation of the relative velocity (derivation Proposition 9.1, §9.3). With the stiffness present it is not proven second order; U2-17 and V-B1 measure it, and U2-18 checks the stability of both choices of F3. The fallback is an element-local exponential integrator for the drag, still inside `multiphysics/`.
2. **Fluid penalty.** Coercivity with (6.4)–(6.5) is inferred, not proven; U2-5 fixes κ_w with a margin of 2, and U2-18 measures its effect on the time step.
3. **Divergence-free w modes** have zero stiffness, as in the continuum. The drag damps them; U2-5 counts them and U2-16 watches them.
4. **Cost.** Twice the unknowns, and the Z blocks run on plain assembled matrices while the u block keeps today's fast paths. Inferred 2–4 times the elastic step until P6; measured in P2.6 on the Mac before any Vista run is sized.

5. **Fault-pressure resolution.** The across-fault pressure boundary layer is metres thick in `tpv5-poro` (diffusivity 3.3 m²/s) and unresolved, in both codes. Dunham & Rice (2008) treat it analytically; a boundary-layer model for p_f is a possible later extension, not part of this plan.
6. **SeisSol version.** Vista's checkout is `31f40bb`, the record was made at `3fc8c627`. S0 decides whether our build reproduces the record; if not, building at `3fc8c627` is the next question for Chunhui Zhao.
7. **Small signals in S1.** Without a fault the starting pressure only diffuses (2.6 m in 2 s) and radiates weak waves, so S1 compares small numbers; both codes run in double precision, and the pass test is relative.
8. **Periodic meshes.** If MFEM's periodic meshes do not work with our integrators, F-D uses exact boundary data, as the elastic plane-wave tests do (T-09(a)).
9. **Splitting error in quasi-dynamic.** Nucleation at intermediate step sizes is the place to watch; V-Q6 and U7-6 measure it.
10. **Implicit solve cost.** The coupled system has twice the unknowns of today's elliptic solve and changes with Δt; D8 keeps factorizations rare, and the cost is measured on the BP5 mesh before any long run.
11. **Allocations.** EAR26024 had 72 SU at the last check and expires on 30 Nov; it is rechecked before every submission.
12. **Branch logistics.** The quasi-dynamic speedups and the fully dynamic cycle live on other branches. P7 starts from trunk; bringing either branch in is Chunhui Zhao's call at that time, with a review.

9. Runs, costs and approvals

Costs are inferred, not measured, and could be off by a factor of 2. The basis: the elastic TPV205 200 m $p = 2$ gold (job 1075916, 16 nodes) took about 4.0 SU for 12 s before the speedups; trunk steps are 2.91 times faster (job 1079640), while setup and output are not, which puts a trunk run at about 1.5 SU. The poroelastic step is taken as 3 times the elastic one (risk 4), so 12 s of the undrained-equivalent TPV205 costs about 4.4 SU. `tpv5-poro`'s slower waves allow a step 1.67 times longer, so 2 s at 200 m costs about 0.4 SU at $p = 2$, 1.9 SU at $p = 3$ (4.5 times) and 0.05 SU at $p = 1$; at 100 m, $p = 2$ costs about 14 times the 200 m run (2.9 times the elements, 5 times the steps). Each run adds about 0.2 SU of setup. Vista bills 0.33 SU per node-hour. Every number is replaced by the measured one once the first run of its kind finishes.

Item	Where	Estimated cost	The words that approve it
F2, F3, D9–D12	—	—	"decisions as recommended", or each answer
Derivation rev 1.2 and this plan	—	—	"approve the derivation rev 1.2 and plan rev 3"
Trunk merge (P0.0)	this thread	—	"approve the trunk merge into the poroelastic branch", after its review page
Each step's commit	this thread	—	"approve step P*n.m*", after its review page

Mac session for the unit tests	Mac	—	the card that starts it; it builds and tests in `/Users/chunhuizhao/projects/seas-mfem-poroelastic` only
SeisSol (4 CPU builds) and PUMGen builds; mesh conversions	Vista, EAR26024	about 2 SU	"submit the SeisSol and PUMGen builds", after step P5.1 is pushed
S0	Vista, EAR26024	under 0.5 SU	"submit S0"
S2 at 1000 m, SeisSol (cost probe)	Vista, EAR26024	under 1 SU	"submit the 1000 m probe"
V-F1 (b), 200 m $p = 2$, 12 s	Vista, EAR26024	about 5 SU	"submit V-F1"
S1, ours, $p = 1$–3	Vista, EAR26024	about 3 SU	"submit S1"
S2 at 200 m, ours, $p = 1$–3, plus one damped $p = 2$ run	Vista, EAR26024	about 4 SU	"submit S2 at 200 m"
S1 and S2 at 200 m, SeisSol	Vista, EAR26024	sized by the probe; any run over 10 SU comes back first	included in "submit S1" and "submit S2 at 200 m"
S2 at 100 m, both codes	Vista, EAR24034 (D10)	ours about 8 SU; SeisSol sized by the probe, any run over 40 SU comes back first	"submit S2 at 100 m"
S3, S4	Vista	sized after S2	asked after S2

Our EAR26024 items add up to about 16 SU, plus SeisSol's share, against the 72 SU of the last check, which the other benchmark threads also draw on. Every Vista job runs from a pushed commit with a clean tree, starts with the CLAUDE.md guard and records the commit in its output, and is submitted only after the words in its row; one row's words never cover another row. The Vista checkout `$WORK/seas-mfem-poroelastic` is clean at `5b2c77d` and not built; the new SeisSol build folders and the PUMGen clone go under `$WORK/seissol` beside the existing builds, which are left untouched. Results that become references get a gold folder with a README (commit, job id, script, deck, mesh sha256, build and run commands), data, comparison figures and speed, as for the elastic golds.

10. Where it lives

- This plan: `/mnt/project-files/poroelastic/plan-2026-10-20/poro-plan-rev3.pdf`, source beside it, and the reference script `poro_test_refs.py` with `refs.json`. Revisions 1–2.2 stay

in `plan-2026-10-19/`.

- The derivation: `/mnt/project-files/poroelastic/derivation-2026-10-20/poro-derivation-rev1.2.pdf`.
- Literature notes and the `tpv5-poro` record: `/mnt/project-files/literature/poroelastic/`; the two receiver files used by U5-1 (`tpv-faultreceivers.h5`, `tpv-receivers.h5`) are in `tpv5-poro/precomputed_double/` (copied 19 Oct; on 20 Oct their sha256 matched the files in SeisSol's `precomputed/double/`).
- Code: branch `nitsche-dynamic-rupture-poroelastic`, folder `miniapps/seas/multiphysics/poroelastic/`; the Mac checkout `/Users/chunhuizhao/projects/seas-mfem-poroelastic`; the Vista checkout `$WORK/seas-mfem-poroelastic`.